

Characteristic Response of a Photomultiplier Tube (PMT)

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The characteristics of a photomultiplier tube (PMT) were studied by analyzing its gain, spectral response, and dark current behavior. The gain was measured as a function of applied voltage and was found to follow a power-law dependence, yielding a parameter $\alpha n = 4.254 \pm 0.22$. The spectral response was evaluated using optical filters in the range 400–700 nm, showing maximum sensitivity in the visible region. The dark current was observed to increase significantly at higher control voltages. The results are consistent with the expected performance of a PMT and demonstrate its effectiveness as a high-gain photodetector.

I. OBJECTIVE

1. Study the spectral response of a PMT using optical filters.
2. Evaluate the current gain of the PMT
3. Study the variation of Dark current with control voltage in a PMT.

II. THEORY

A Photomultiplier Tube (PMT) is a sensitive photodetector used for the detection of low-intensity light in the ultraviolet (UV), visible, and near-infrared regions. It is widely used in nuclear and particle physics experiments due to its high gain, fast response, and low noise characteristics.

A. Working Principle

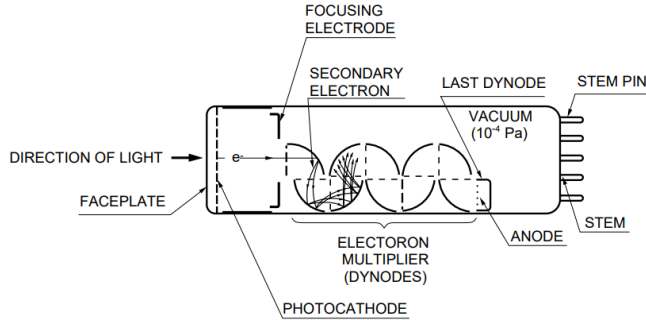


FIG. 1. Construction of a typical PMT.

When light enters the PMT, incident photons strike the photocathode, resulting in emission of photoelectrons

via the *photoelectric effect*. These photoelectrons are accelerated and directed towards the first dynode by focusing electrodes. At each dynode, secondary electron emission occurs, multiplying the number of electrons (see Figure 1). This multiplication process continues through successive dynodes, forming an electron avalanche. The amplified electron signal is finally collected at the anode, producing an output signal.

B. Gain

One of the most important property of a PMT is its ability to amplify a small photocurrent into a measurable signal. If δ is the average secondary emission ratio (number of electrons emitted per incident electron) and n is the number of dynode stages, the total current gain μ is given by [1]:

$$\mu = \delta^n \quad (1)$$

The secondary emission ratio depends on the interdynode voltage E and is given by:

$$\delta = AE^\alpha \quad (2)$$

where, A is a constant, E is the interstage voltage, α is a material-dependent constant ($\sim 0.7 - 0.8$).

For an applied voltage V across the PMT, the gain becomes:

$$\mu = (AE^\alpha)^n = KV^{\alpha n} \quad (3)$$

where K is a constant.

Thus, the gain depends strongly on the applied voltage.

Experimentally, instead of current gain, voltage gain G is often measured:

$$G = \frac{\text{Output Voltage}}{\text{Input Voltage}} = \frac{V_{out}}{V_{in}} \quad (4)$$

For two different applied voltages V_1 and V_2 , the ratio of gains is given by:

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$$\frac{G_2}{G_1} = K \left(\frac{V_2}{V_1} \right)^{\alpha n} \quad (5)$$

This relation is used to experimentally determine the exponent αn using suitable graphs.

C. Spectral Response

The photocathode converts incident photons into photoelectrons with an efficiency that depends on wavelength. This variation is called the *spectral response*.

It is determined by the photocathode material (long wavelength limit) and the window material (short wavelength limit). The spectral response is typically expressed in terms of sensitivity as a function of wavelength.

D. Dark Current

Even in complete darkness, a small current flows in the PMT known as *dark current*. This contributes to noise and limits the detection capability.

Major sources of dark current include:

- Thermionic emission
- Ionization of residual gases
- Glass scintillation
- Leakage currents
- Field emission

Dark current strongly depends on the applied voltage.



FIG. 2. The Hamamatsu H9305-03 PMT module used in the experiment.

E. PMT Module Used in Experiment

In this experiment, a PMT module is used which integrates photomultiplier tube, high-voltage supply and voltage divider network.



FIG. 3. External power supply and control voltage unit used to operate the PMT module.

The module used in this experiment is Hamamatsu H9305-03 PMT module, which contains a 13-mm (half inch) diameter side-on PMT in a compact aluminum housing. It has a reflection mode multialkali (Na-K-Sb-Cs) photocathode, an adequate gain of more than 10^6 .

Figure 3 shows an external power supply of $\pm 15V$ which provides the driving voltage as well as the control voltage for the PMT module. The control voltage can be varied from +0.25 to 1V using a potentiometer. The ratio of the control voltage to actual PMT voltage is $1 : 10^3$.

The PMT serves as a highly sensitive device where weak optical signals are converted into measurable electrical signals through electron multiplication.

III. APPARATUS REQUIRED

1. PMT Module (H9305-03) (see Figure 2).
2. Control voltage unit
3. Digital Storage Oscilloscope (DSO)
4. Tungsten filament bulb
5. DC power supply
6. Optical filters for different wavelengths (400–700 nm) with suitable holder
7. Mounted photodiode
8. Multimeters and connecting wires

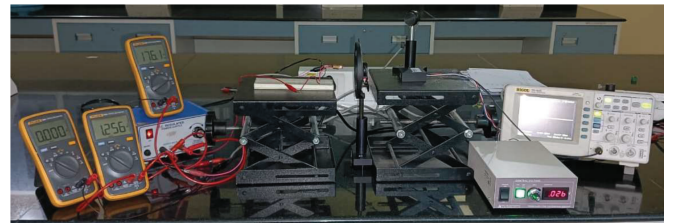


FIG. 4. Experimental setup for measuring the gain and spectral response of the PMT using a light source, optical filters, photodiode, and DSO.

IV. OBSERVATIONS

TABLE I. Experimental observations with photodiode and PMT. Notation: λ = wavelength, V_S = supply voltage, I_S = supply current, $\mathcal{R}$ = radiant sensitivity ($I_{\text{PMT}}/P_{\text{PD}}$), where I_{PMT} is PMT Output Current and P_{PD} is Output power of Photodiode.

λ (nm)	V_S (V)	I_S (mA)	Photodiode			PMT		$\mathcal{R}$ (A/W)
			Voltage (V)	Current (μ A)	Power (μ W)	Voltage (V)	Current (μ A)	
500	4.40	141.0	0.194	0.1	0.0194	4.40	4.40	226.80
635	3.245	119.8	0.194	0.1	0.0194	3.12	3.12	160.82
570	3.279	120.5	0.194	0.1	0.0194	2.80	2.80	144.33
460	5.42	157.6	0.193	0.1	0.0193	4.80	4.80	248.70
540	3.344	121.6	0.194	0.1	0.0194	3.36	3.36	173.20

TABLE II. PMT Gain Characteristics (PMT Input Voltage = 0.194 V)

Supply Voltage (V)	PMT Output (V)	Gain	$\log_{10}(G)$	$\log_{10}(V_{\text{supply}})$
0.300	0.38	1.27	0.1026	-0.5229
0.350	1.20	3.43	0.5351	-0.4559
0.400	2.30	5.75	0.7597	-0.3979
0.450	6.20	13.78	1.1392	-0.3468
0.500	11.80	23.60	1.3729	-0.3010
0.550	14.60	26.55	1.4240	-0.2596
0.600	18.00	30.00	1.4771	-0.2218
0.650	28.80	44.31	1.6465	-0.1871
0.700	42.80	61.14	1.7864	-0.1549
0.750	58.80	78.40	1.8943	-0.1249
0.800	84.40	105.50	2.0233	-0.0969

TABLE III. Observations for dark current. (V_C = Control voltage, V_{PMT} = PMT output voltage)

V_C (V)	V_{PMT} (V)	Dark Current (nA)
0.30	0.000	0
0.40	0.000	0
0.50	0.000	0
0.60	0.000	0
0.70	0.001	0.1
0.75	0.003	0.3
0.80	0.006	0.6
0.85	0.010	1.0
0.90	0.017	1.7
0.95	0.029	2.9
1.00	0.042	4.2

V. DATA ANALYSIS

A. Spectral Response

The spectral response of the PMT was studied using optical filters of different wavelengths in the range 400–700 nm. The input optical power was estimated using a photodiode, and the PMT output current was measured.

The radiant sensitivity $\mathcal{R}$ was calculated using:

$$\mathcal{R} = \text{radiant sensitivity} = \frac{I_{\text{PMT}}}{P_{\text{PD}}} \quad (6)$$

where I_{PMT} is PMT Output Current and P_{PD} is Output power of Photodiode.

A plot of radiant sensitivity versus wavelength was obtained. The variation of sensitivity with wavelength reflects the dependence of photocathode efficiency on photon energy. The observed trend is consistent with the

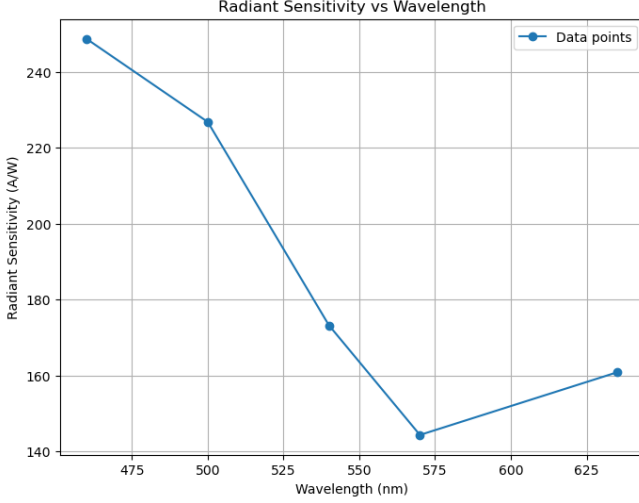


FIG. 5. Spectral response of the PMT showing variation of radiant sensitivity with wavelength in the range 400–700 nm.

expected spectral response of a multialkali photocathode, showing higher sensitivity in the visible region. The reading for I_{PD} are equal to the least count of the micrometer (i.e. $0.1\mu A$), due to which on error propagation, we obtain more than 100% error on individual readings. That's why we did not include error bars on individual readings in the Figure 5.

B. Gain

The gain of the PMT was evaluated by measuring the output voltage for a fixed input signal while varying the control (supply) voltage. The voltage gain was calculated using Eq. (4). Using the relation:

$$G \propto V^{\alpha n} \quad (7)$$

Taking logarithm on both sides,

$$\log G = (\alpha n) \log V + \log K \quad (8)$$

A linear fit plot of $\log G$ versus $\log V$ was obtained and fitted to a straight line (see Figure 6).

We fit the observed data-points $\{x_i, y_i\}$ into a form,

$$y = mx + c \quad (9)$$

The statistical error calculated in linear fit is done using the following formulae [2],

$$\sigma_y = \sqrt{\left(\sum_i \frac{(y_n - y_i)^2}{N - 2}\right)} \quad (10)$$

Errors in slope and intercept is given by,

$$\Delta m = \sigma_y \sqrt{\frac{\sum_i x_i^2}{((N \sum_i x_i^2) - (\sum_i x_i)^2)}} \quad (11)$$

$$\Delta c = \sigma_y \sqrt{\frac{N}{((N \sum_i x_i^2) - (\sum_i x_i)^2)}} \quad (12)$$

The slope of this graph gives the value of αn . We obtain after linear fit,

$$\alpha n = \text{slope} = 4.254$$

$$\Delta(\alpha n) = \Delta(\text{slope}) = 0.22$$

This characterizes the multiplication process inside the PMT.

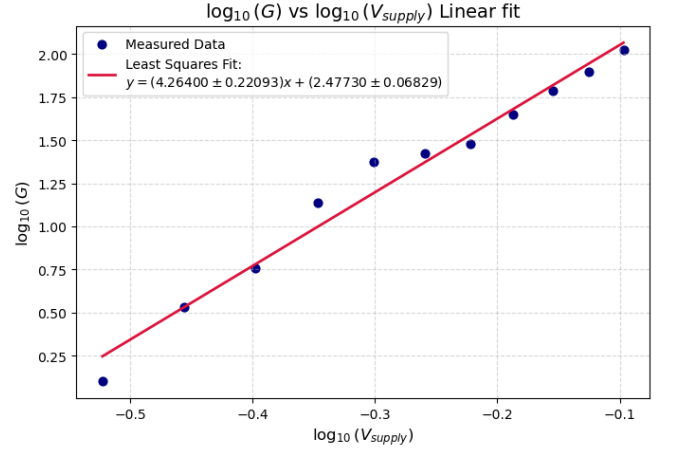


FIG. 6. Log-log plot of PMT gain versus applied voltage with linear fit used to determine the parameter αn .

C. Dark Current

The dark current was measured by completely shielding the PMT from external light and recording the output current as a function of control voltage.

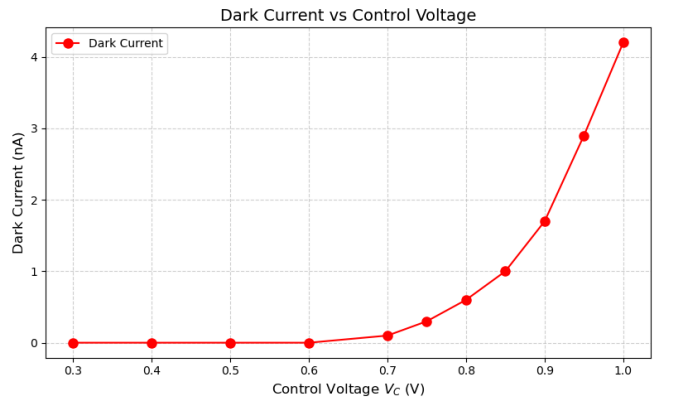


FIG. 7. Variation of dark current with control voltage, illustrating the increase in noise at higher operating voltages.

The results show that the dark current remains negligible at low voltages and increases significantly beyond

a certain threshold voltage. This behavior is attributed to increased thermionic emission and field emission at higher voltages.

A plot of dark current vs control voltage shows a non-linear increase, indicating that noise contributions become more significant at higher operating voltages (see Figure 7). This limits the usable operating range of the PMT for low-light measurements.

All the plots, data files and linear fit code are available on this GitHub Repository [3].

VI. RESULTS AND DISCUSSION

The gain characteristics of the PMT were studied by plotting $\log G$ vs $\log V$. A linear fit to the data yielded:

$$\alpha n = 4.254 \pm 0.22 \quad (13)$$

This confirms the expected power-law dependence of gain on applied voltage. The relatively high value of αn indicates strong electron multiplication across the dynode stages, consistent with the theoretical model.

The spectral response measurements show that the radiant sensitivity varies with wavelength, with higher sensitivity observed in the visible region. The maximum response occurs around shorter visible wavelengths, which is characteristic of a multialkali photocathode. Minor deviations from the expected trend may arise due to filter imperfections and alignment errors.

The dark current measurements indicate that the current remains negligible at low control voltages and increases nonlinearly beyond 0.7 V. This behavior is attributed to thermionic emission and field emission effects. The rapid increase at higher voltages highlights

the trade-off between achieving high gain and minimizing noise.

Overall, the experimental observations are in good agreement with the theoretical expectations for a photomultiplier tube.

VII. CONCLUSION

In this experiment, the characteristics of a photomultiplier tube were successfully studied. The gain of the PMT was found to follow a power-law dependence on the applied voltage, with the parameter:

$$\alpha n = 4.254 \pm 0.22 \quad (14)$$

The spectral response analysis confirmed that the PMT exhibits maximum sensitivity in the visible wavelength range, consistent with the properties of the photocathode material. The dark current was observed to increase significantly at higher control voltages, limiting the optimal operating range of the device.

Thus, the experiment verifies the fundamental working principles of the PMT and demonstrates its effectiveness as a high-gain, low-light detection device.

VIII. SOURCES OF ERROR

- Fluctuations in the intensity of light leading to uncertainty in input signal.
- Limited precision of multimeters used for current and voltage measurements.

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- [3] Aryan Shrivastava. P341 - nuclear physics and instrumentation lab. <https://github.com/crimsonpane23/P341-Nuclear-Physics-and-Instrumentation-Lab.git>, 2026.